Biochemistry Cume
October 2, 2009

Total points: 100

1. In the monosaccharide derivatives known as sugar alcohols, the carbonyl oxygen is reduced to a hydroxyl group. For example, D-glyceraldehyde can be reduced to glycerol. However, this sugar alcohol is no longer designated D or L. Why not? (**10 points**)

2. Although lactose exists in two anomeric forms, no anomeric forms of sucrose have been reported. Why not? (**10 points**)

3. Fructose in honey is mainly in the β-D-pyranose form. This is one of the sweetest substances known, about twice as sweet as glucose. The β-D-furanose form of fructose is much less sweet. The sweetness of honey gradually decreases at a high temperature. Also, high fructose corn syrup (a commercial product in which much of glucose in corn syrup is converted to fructose) is used for sweetening cold but not hot drinks. What chemical property of fructose could account for these observations? Explain. (**20 points**)

4. Hydrolysis of sucrose (specific rotation = $+66.5^{o}$) yields an equimolar mixture of D-glucose (specific rotation = 52.5^{o}) and D-fructose (specific rotation = -92°).

   (**60 points**)

A. Suggest a convenient way to determine the rate of hydrolysis of sucrose by an enzyme preparation extracted from the lining of the small intestine.

B. Explain why equimolar amounts of D-glucose and D-fructose formed by hydrolysis of sucrose are called invert sugar in the food industry.

C. The enzyme *invertase* is allowed to act on a solution of sucrose until the optical rotation of the solution becomes zero. Calculate the fraction of sucrose hydrolyzed.